

FashionHub Design Document

Project Description

FashionHub is a web-based internal management system designed for small women's clothing brands to manage their daily business operations more efficiently. Many small fashion businesses rely on spreadsheets, chat messages, or scattered documents to store operational information, which often results in inconsistent data and reduced productivity. FashionHub addresses this problem by integrating product management, employee management, and supplier management into a single centralized platform that is clean, fast, and easy to use.

The system is developed using Node.js and Express for the backend, MongoDB as the database using the native MongoDB driver, and vanilla HTML, CSS, and JavaScript for the frontend with full client-side rendering. The application also includes authentication to protect access and a dashboard to provide business insights.

System Overview

FashionHub combines multiple independent modules into one application. The product catalog module allows users to manage clothing items, track inventory, update pricing, and monitor stock levels. The employee management module provides functionality for maintaining staff records, including departments, roles, salaries, and employment status. The supplier management module organizes vendor information, contact details, and performance ratings. A dashboard provides a quick overview of business data, including statistics and alerts.

The system follows a modular architecture with a RESTful API backend and a client-side rendered frontend that communicates using HTTP requests.

User Personas

One primary user of the system is Lily, a product manager responsible for maintaining hundreds of clothing products. She needs a centralized platform to update product information, track inventory levels, and monitor low-stock alerts so that she can ensure smooth business operations. Without such a system, she faces challenges with inconsistent data across spreadsheets and communication tools.

Another important user is Emma, a procurement specialist who manages supplier relationships. She needs a reliable way to organize vendor contacts, compare supplier

performance, and quickly retrieve supplier information. Her current workflow involves searching through emails and messages, which is inefficient and prone to errors.

A third user is Sophie, the brand owner who oversees business operations. She needs a dashboard that provides a quick overview of inventory status, employee distribution, and supplier performance. Her goal is to make informed decisions based on accurate and accessible information.

User Stories

The system supports several key user interactions. A user should be able to register and log in so that access to the system is restricted to authorized individuals. A product manager should be able to create, update, and delete product records to maintain accurate inventory data. The system should also provide alerts when inventory levels fall below a threshold so that restocking decisions can be made in time.

An operations or HR manager should be able to create and maintain employee records, update employee details when changes occur, and remove inactive employees to keep the database accurate. A procurement specialist should be able to manage supplier information, update contact details, and assign performance ratings so that vendor relationships can be evaluated over time.

MVP Scope

The minimum viable product includes user authentication with registration and login functionality, full CRUD operations for products, employees, and suppliers, a dashboard displaying business statistics, and client-side filtering and rendering features. The application also includes seeded data to demonstrate realistic usage scenarios.

System Architecture

The frontend is built using standard HTML, CSS, and vanilla JavaScript with client-side rendering logic implemented through the Fetch API. The backend is developed using Node.js with Express and follows a modular route structure to separate concerns. MongoDB Atlas is used as the database, accessed through the native MongoDB driver rather than an object modeling library. The application is deployed on a cloud hosting platform to allow public access.

Work Distribution

The project is developed by two team members with clearly defined responsibilities. Xuyang Shen is responsible for the authentication system, the product catalog module, the employee management module, and the dashboard with business statistics and alerts. Gaoyuan Shi is responsible for the supplier management module, including supplier CRUD functionality, filtering, search capabilities, and the rating and notes system.

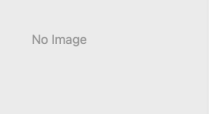
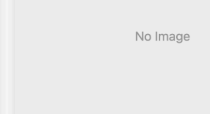
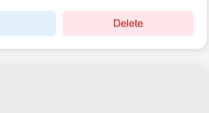
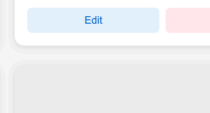
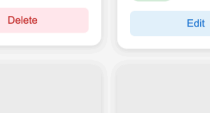
Design Mockups

The user interface follows a clean dashboard-oriented layout that allows users to navigate between modules easily. The login page provides authentication functionality for secure access. The dashboard page presents business statistics and quick summaries. The products page allows users to manage product information through forms and tables. The employees page provides functionality for maintaining employee records. The suppliers page allows users to manage supplier relationships, including ratings and notes. Screenshots of these pages are included below to illustrate the interface design.

Here's your business overview for today.

 Low Stock Alerts View All Products →	
Classic Navy Maxi Dress	2 in stock
Modern Yellow Leggings	2 in stock
Classic Pink Shift Dress	5 in stock
Chic Navy Polo Shirt	4 in stock
Luxury Beige Leggings	3 in stock
Luxury Teal Sun Dress	9 in stock
Cozy Green Culottes	1 in stock
Sporty Teal Culottes	4 in stock
Sporty Green Blazer	3 in stock

+ Add Product

 No Image Luxury Black Mini Dress \$168.79 dresses Stock: 199 active Edit Delete	 No Image Luxury Brown Palazzo Pants \$60.33 bottoms Stock: 197 active Edit Delete	 No Image Bohemian Black Sweater \$18.43 tops Stock: 358 active Edit Delete
 No Image Cozy Navy Hoodie \$191.79	 No Image Relaxed Green Shift Dress \$143.51	 No Image Elegant Pink Culottes \$40.46

Employee Management							Dashboard	Products	Employees	Suppliers	Logout
All Departments 							+ Add Employee				
Name	Email	Department	Position	Hire Date	Status	Actions					
Aria Rodriguez	aria.rodriguez443@fashionhub.com	production	Machine Operator	2025-02-19	active	Edit Delete					
Ella Martin	ella.martin688@fashionhub.com	hr	Training Coordinator	2023-06-12	active	Edit Delete					
Alexander Miller	alexander.miller50@fashionhub.com	sales	Sales Manager	2021-06-11	inactive	Edit Delete					
Sophia Thomas	sophia.thomas93@fashionhub.com	hr	Payroll Specialist	2020-04-30	active	Edit Delete					
Evelyn Moore	evelyn.moore797@fashionhub.com	production	Production Manager	2023-07-14	active	Edit Delete					
Ella Harris	ella.harris822@fashionhub.com	design	Pattern Maker	2024-12-13	active	Edit Delete					
James White	james.white786@fashionhub.com	production	Quality Control	2024-11-30	active	Edit Delete					

Supplier Management							Dashboard	Products	Employees	Suppliers	Logout
All Categories Search suppliers...							+ Add Supplier				
Company	Contact	Phone	Email	Categories	Rating	Notes	Actions				
Harmony Textiles #5 2263 Market St, Miami	Lily Smith	(356) 652-9675	contact98@harmonytextiles.com	accessories, manufacturing, fabrics	1 ★	Preferred vendor	Edit Delete				
Silk Road Textiles #9 7871 Broadway, Miami	Chloe Brown	(308) 505-1598	contact160@silkroadtextiles.com	fabrics, accessories, packaging	2 ★	Preferred vendor	Edit Delete				
Atlas Accessories #6 9426 Pine Rd, Dallas	Amelia Moore	(586) 420-4766	contact116@atlasaccessories.com	fabrics	3 ★	Bulk discount available	Edit Delete				
Pioneer Fabrics #4 2168 Market St, Seattle	Mia Taylor	(353) 965-1417	contact75@pioneerfabrics.com	fabrics, accessories	5 ★	Good customer service	Edit Delete				
Dragon Textiles #12 4171 Market St, Dallas	Charlotte Anderson	(674) 309-9299	contact225@dragontextiles.com	fabrics, accessories, packaging	1 ★	Bulk discount available	Edit Delete				
Silk Road Textiles #2 3322 Pine Rd, Portland	Layla Moore	(733) 313-2514	contact20@silkroadtextiles.com	manufacturing, fabrics, accessories	5 ★	Reliable delivery times	Edit Delete				
Atlas Accessories #8	Harper	(920)				Bulk discount	Edit				

FashionHub

Product Catalog Management

Login

Register

Username

Ujang

Password

Login